

Q1 (2)

$$\sqrt{3}kx + ky = 4\sqrt{3}$$

$$\sqrt{3}kx - ky = 4\sqrt{3}k^2$$

Adding equation (1) & (2) $2\sqrt{3}kx = 4\sqrt{3}(k^2 + 1)$

$$x = 2\left(k + \frac{1}{k}\right)$$

Subtracting equation (1) & (2) $y = 2\sqrt{3}\left(\frac{1}{k} - k\right)$

$$\therefore \frac{x^2}{4} - \frac{y^2}{12} = 4$$

$$\frac{x^2}{16} - \frac{y^2}{48} = 1 \quad \text{Hyperbola}$$

$$\therefore e^2 = 1 + \frac{48}{16}$$

$$e = 2$$

Q2 (2)

$$e_1 = \sqrt{1 - \frac{16}{25}} = \frac{3}{5} \quad \text{foci } (\pm ae, 0)$$

$$\text{Foci} = (\pm 3, 0)$$

Let equation of hyperbolabe $\frac{x^2}{A^2} - \frac{y^2}{B^2} = 1$ Passes through $(\pm 3, 0)$ $A^2 = 9$, $A = 3$, $e_2 = \frac{5}{3}$

$$e_2^2 = 1 + \frac{B^2}{A^2}$$

$$\frac{25}{9} = 1 + \frac{B^2}{9} \Rightarrow B^2 = 16$$

$$\text{Ans } \frac{x^2}{9} - \frac{y^2}{16} = 1$$

Q1: 25 Feb (Shift 1) - Numerical

The locus of the point of intersection of the lines $(\sqrt{3})kx + ky - 4\sqrt{3} = 0$ and $\sqrt{3}x - y - 4(\sqrt{3})k = 0$ is a conic, whose eccentricity is

Q2: 25 Feb (Shift 2) - Single Correct

A hyperbola passes through the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and its transverse and conjugate axes coincide with major and minor axes of the ellipse, respectively. If the product of their eccentricities is one, then the equation of the hyperbola is:

(1) $\frac{x^2}{9} - \frac{y^2}{4} = 1$

(2) $\frac{x^2}{9} - \frac{y^2}{16} = 1$

(3) $x^2 - y^2 = 9$

(4) $\frac{x^2}{9} - \frac{y^2}{25} = 1$

